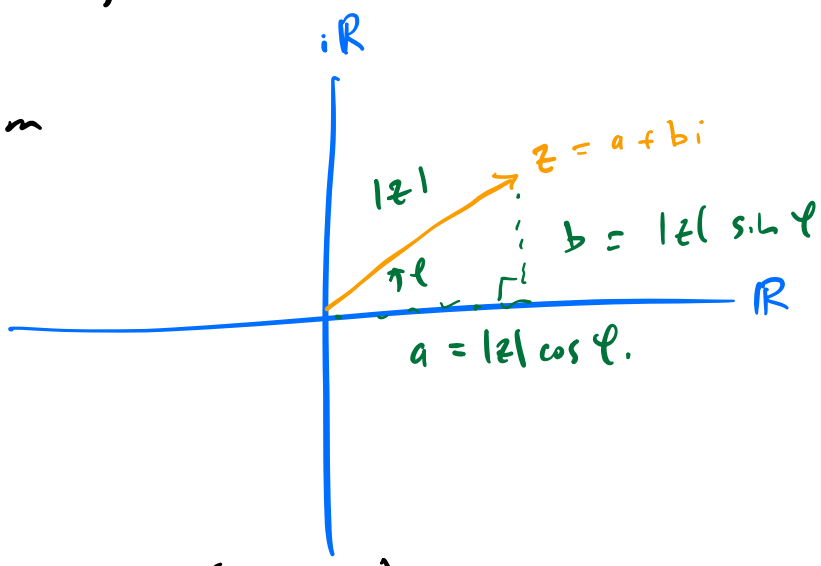
Complex numbers

$\mathbb{C} = \{a+bi : a,b \in \mathbb{R}\}$

Polar form



$$z = |z|\cos\varphi + (|z|\sin\varphi)i$$
$$= |z|\, e^{i\varphi}, \quad \text{where } e^{i\varphi} = \cos\varphi + i\sin\varphi.$$

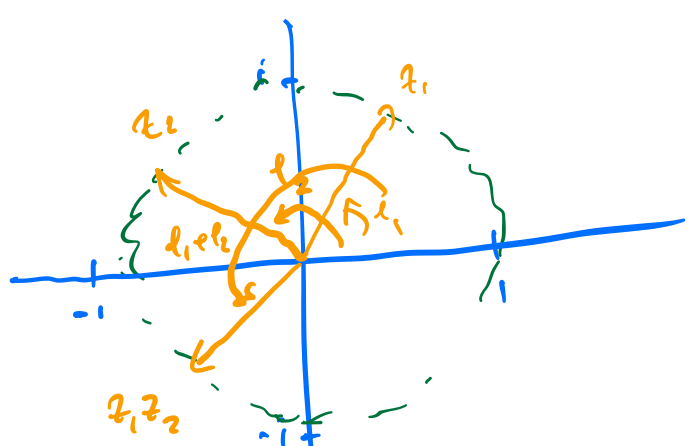
φ = argument of z, can be taken with $0 \le \varphi < 2\pi$

Polar form of z is $re^{i\varphi}$ where $r = |z| > 0$ and φ is angle from positive real axis.

Theorem

Suppose $z_1 = r_1 e^{i\varphi_1}$ and $z_2 = r_1 e^{i\varphi_2}$ are non zero complex numbers. Then

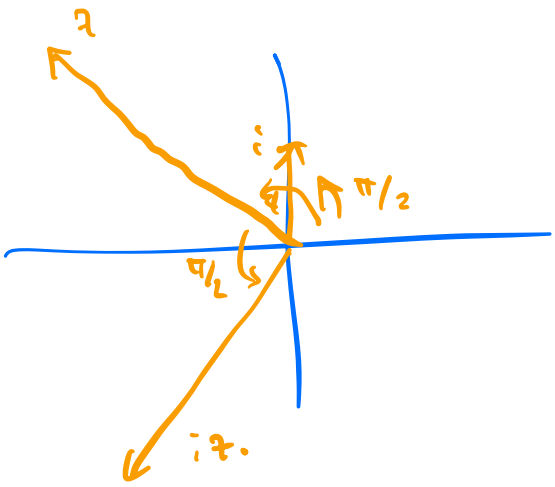
$$z_1 z_2 = (r_1 r_2)\, e^{i(\varphi_1+\varphi_2)}$$



$z_1 z_2 = e^{i(\varphi_1+\varphi_2)}$

$z = re^{i\varphi}$

$iz = re^{i(\varphi+\pi/2)}$



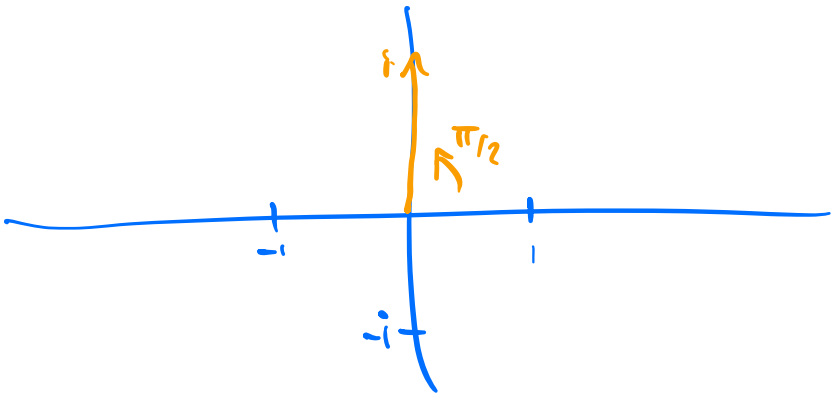
Theorem [De Moivre]

Suppose $z = re^{i\varphi}$ in polar form.

Then $z^k = r^k e^{ik\varphi}$.

For example

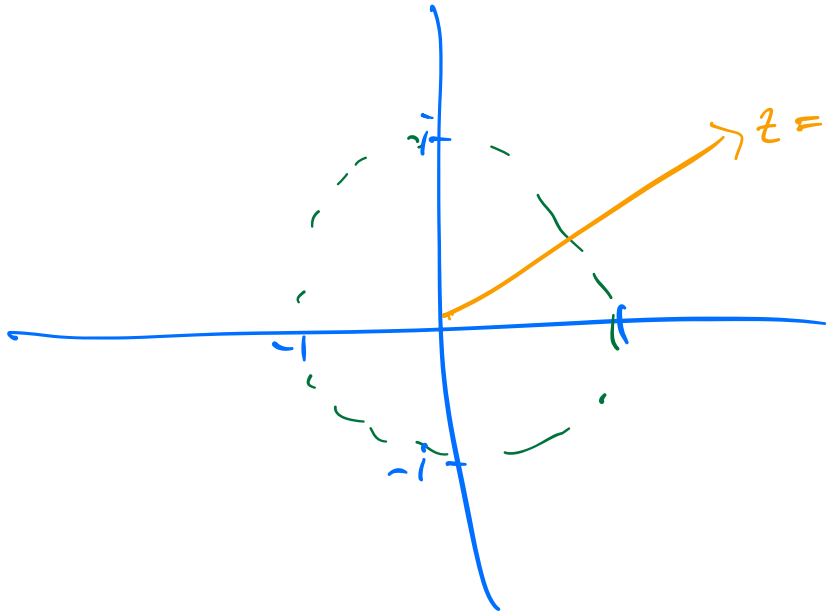
$$re^{i\varphi} re^{i\varphi} = r^2 e^{i\varphi} e^{i\varphi} = r^2 e^{i2\varphi}$$

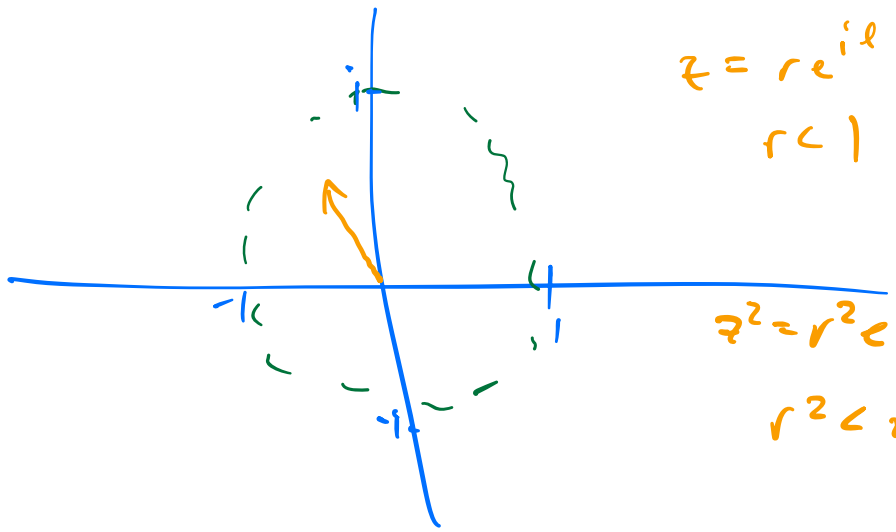


$i = e^{i\pi/2}$, $\quad i^2 = e^{i2(\pi/2)} = e^{i\pi} = 1$

$i^3 = e^{i3\pi/2} = -i$, $\quad i^4 = e^{i4\pi/2} = e^{2\pi i} = 1$

$$i^k = \begin{cases} i & k \text{ remainder } 1 \bmod 4 \\ -1 & k \text{ remainder } 2 \bmod 4 \\ -i & k \text{ remainder } 3 \bmod 4 \\ 1 & k \text{ remainder } 0 \bmod 4 \end{cases}$$





Theorem

Suppose A and B are complex matrices which can be multiplied. Then

$$\overline{AB} = \overline{A}\,\overline{B}.$$

Here $\overline{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \overline{a_{11}} & \cdots & \overline{a_{1n}} \\ \overline{a_{m1}} & & \overline{a_{mn}} \end{bmatrix}$.

Example

$$\begin{bmatrix} 1+i & i \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1-i & -2i \\ i & -i \end{bmatrix} = \begin{bmatrix} 1-i & -i \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1+i & 2i \\ -i & i \end{bmatrix}$$

5.5 Complex eigenvalues

Theorem Everything we've done for real vector spaces also holds for complex vector spaces.

e.g. $\mathbb{C}^n$, $M_{m\times n}(\mathbb{C})$, complex-valued functions

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Example

Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in M_{2\times 2}(\mathbb{C})$

$\det(A - xI) = \det\left(\begin{bmatrix} -x & 1 \\ -1 & -x \end{bmatrix}\right) = x^2+1$

has two roots $\lambda_1 = i$ and $\lambda_2 = -i$.

$\Rightarrow \det(A - xI) = (i-x)(-i-x) = (x-i)(x+i)$

A has two distinct complex eigenvalues, i and $-i$.

Let's diagonalize A.

To compute the i-eigenspace, we compute

$$\mathrm{Nul}(A - iI) = \mathrm{Nul}\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}\right)$$

$$= \mathrm{Nul}\left(\begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix}\right)$$

Row reduce $\begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \rightsquigarrow \begin{bmatrix} -1 & -i \\ -i & 1 \end{bmatrix}$ $(R_1 \leftrightarrow R_2)$

$$\rightsquigarrow \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \quad ((-1)R_1)$$

$$\rightsquigarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix} \quad (iR + R_2)$$

$iR_1 + R_2$

$i\begin{bmatrix} 1 & i \end{bmatrix} + \begin{bmatrix} -i & 1 \end{bmatrix}$

$= \begin{bmatrix} i & -1 \end{bmatrix} + \begin{bmatrix} -i & 1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \end{bmatrix}$

$$\mathrm{Nul}(A - iI) = \left\{ c_2 \begin{bmatrix} -i \\ 1 \end{bmatrix} : c_2 \in \mathbb{C} \right\}$$

$$= \mathrm{span}\left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$$

Similarly the $-i$-eigenspace is

$$\mathrm{Nul}(A + iI) = \dots = \mathrm{span}\left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$$

Finally $A = \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix}^{-1}$

Theorem

Suppose A is an $n \times n$ matrix with <u>real</u> entries, and $\lambda \in \mathbb{C}$ is an eigenvalue with eigenvector $v \in \mathbb{C}^n$. Then $\bar{\lambda} \in \mathbb{C}$ is also an eigenvalue with eigenvector $\bar{v} \in \mathbb{C}^n$.

Proof. $Av = \lambda v \Rightarrow \overline{Av} = \overline{\lambda v} \Rightarrow \bar{A}\bar{v} = \bar{\lambda}\bar{v}$

$$\Rightarrow A\bar{v} = \bar{\lambda}\bar{v}$$

Example In example with $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$v = \begin{bmatrix} -i \\ 1 \end{bmatrix}$ is an eigenvector with eigenvalue i

$\bar{v} = \overline{\begin{bmatrix} -i \\ 1 \end{bmatrix}} = \begin{bmatrix} \overline{-i} \\ \bar{1} \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix}$ is an eigenvector

with eigenvalue $\bar{i} = -i$.

Example.

Take $A = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}$. Then

$$\det(A - xI) = \det\left(\begin{bmatrix} 2-x & 3 \\ -3 & 2-x \end{bmatrix}\right) = \cdots = x^2 - 4x + 13$$

Quadratic formula: Roots are $\dfrac{4 \pm \sqrt{(-4)^2 - (4)(13)}}{2}$

$$= 2 \pm \sqrt{-9}$$

$$= 2 \pm 3\sqrt{-1} = 2 \pm 3i$$

Clearly $\overline{2+3i} = 2-3i$.

To diagonalize A we compute $2+3i$ eigenspace

$$\mathrm{Nul}(A-(2+3i)I) = \mathrm{Nul}\left(\begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix} - \begin{bmatrix} 2+3i & 0 \\ 0 & 2+3i \end{bmatrix}\right)$$

$$= \mathrm{Nul}\left(\begin{bmatrix} -3i & 3 \\ -3 & -3i \end{bmatrix}\right)$$

Row reduce $\begin{bmatrix} -3i & 3 \\ -3 & -3i \end{bmatrix} \rightsquigarrow \begin{bmatrix} -3 & -3i \\ -3i & 3 \end{bmatrix}$

$$\rightsquigarrow \begin{bmatrix} -3 & -3i \\ 0 & 0 \end{bmatrix} \quad (-iR_1+R_2)$$

$$\rightsquigarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

$$\mathrm{Nul}(A-(2+3i)I) = \mathrm{span}\left\{\begin{bmatrix} -i \\ 1 \end{bmatrix}\right\} \quad \text{(coincidence)}$$

By previous thm $\mathrm{span}\left\{\overline{\begin{bmatrix} -i \\ 1 \end{bmatrix}}\right\} = \mathrm{span}\left\{\begin{bmatrix} i \\ 1 \end{bmatrix}\right\}$ is

the $\overline{2+3i} = 2-3i$ - eigenspace.

$$\Rightarrow A = \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2+3i & 0 \\ 0 & 2-3i \end{bmatrix} \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix}^{-1}$$

Example.

Take $A = \begin{bmatrix} \frac{1+i}{2} & 1 \\ 0 & 1 \end{bmatrix}$.

$$\det(A - xI) = \left(x - \left(\tfrac{1+i}{2}\right)\right)(x-1)$$

A has two eigenvalues $\lambda_1 = \frac{1+i}{2}$, $\lambda_2 = 1$.

Clearly $\bar{\lambda}_2 = \bar{1} = 1 \neq \lambda_1$

$$\mathrm{Nul}\left(A - \left(\tfrac{1+i}{2}\right)I\right) = \mathrm{Nul}\left(\begin{bmatrix} \frac{1+i}{2} & 1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{1+i}{2} & 0 \\ 0 & \frac{1-i}{2} \end{bmatrix}\right)$$

$$= \mathrm{Nul}\begin{bmatrix} 0 & 1 \\ 0 & 1 - \frac{1+i}{2} \end{bmatrix}$$

$$= \mathrm{Nul}\left(\begin{bmatrix} 0 & 1 \\ 0 & \frac{1-i}{2} \end{bmatrix}\right)$$

Row reduce $\begin{bmatrix} 0 & 1 \\ 0 & \frac{1-i}{2} \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\quad -(1-i/2)R_1 + R_2$

$$\lambda_1\text{-eigenspace} = \mathrm{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\} = \left\{ c \begin{bmatrix} 1 \\ 0 \end{bmatrix} : c \in \mathbb{C} \right\}.$$

$$\subseteq \mathbb{C}^2$$

For the λ_2-eigenspace, we compute

$$\mathrm{Nul}(A-I) = \mathrm{Nul}\left(\begin{bmatrix} \frac{1+i}{2} & 1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)$$

$$= \mathrm{Nul}\left(\begin{bmatrix} \frac{-1+i}{2} & 1 \\ 0 & 0 \end{bmatrix}\right)$$

Row reduce $\begin{bmatrix} \frac{-1+i}{2} & 1 \\ 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -1+i & 2 \\ 0 & 0 \end{bmatrix}$ $2R_1$

$$\rightsquigarrow \begin{bmatrix} 1+1 & 2(-1-i) \\ 0 & 0 \end{bmatrix} (\overline{-1+i})R$$

$z\bar{z} = a^2+b^2$

$z = a+bi$

$$= \begin{bmatrix} 2 & 2(-1-i) \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1-i \\ 0 & 0 \end{bmatrix}$$

$$\mathrm{Nul}(A-I) = \left\{ c_2 \begin{bmatrix} 1+i \\ 1 \end{bmatrix} : c_2 \in \mathbb{C} \right\} = \mathrm{span}\left\{ \begin{bmatrix} 1+i \\ 1 \end{bmatrix} \right\}.$$

$$A = \begin{bmatrix} 1 & 1+i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1+i}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1+i \\ 0 & 1 \end{bmatrix}^{-1}.$$

(Check to see if there is a mistake)